

Computational Linguistics 1

Mini-Assignment 1

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Language Chosen: Hindi

Format:

Original Sentence

Transliteration

Gloss

English

Part A:

1. Lexically Ambiguous Sentences:

Sentence 1:

हार की चमक मेरी आँखें मे बर्करार थी |

Haar kee chamak meri aankhon mein barkaraar thee.

Failure GEN.F sparkle 1.SG.GEN eye.PL.OBL LOC retain AUX.PST.F.SG

The sparkle of defeat remained in my eyes.

हार की चमक मेरी आँखें मे बर्करार थी |

Haar kee chamak meri aankh on mein barkaraar thee.

Necklace GEN.F sparkle 1.SG.GEN eye PL.OBL LOC retain AUX.PST.F.SG

The sparkle of the necklace remained in my eyes.

Sentence 2:

ये आम हैं मगर फिर भी अच्छे हैं |

Ye aam hai magar phir bhee ache hain.

DEM.PL Mango AUX.PRS.PL but still nevertheless good.M.PL AUX.PRS.PL
These are mangoes yet they are still good.

ये आम हैं मगर फिर भी अच्छे हैं |

Ye aam hai magar phir bhee ache hain.

DEM.PL Mango AUX.PRS.PL but still nevertheless good.M.PL AUX.PRS.PL
These are common yet they are still good.

2. Structurally (syntactically) Ambiguous Sentences:

Sentence 3:

वह साहसी कुत्ते का मालिक है |

Wah sahsi kutte ka maalik hai

DEM brave dog.OBL.SG GEN.SG master AUX.PRS.SG

He's the owner of the brave dog.

वह साहसी कुत्ते का मालिक है |

Wah sahsi kutte ka maalik hai

DEM brave dog.OBL.SG GEN.SG master AUX.PRS.SG

He's the brave owner of the dog.

Sentence 4:

राधा और उसकी दोस्त की बिल्लियाँ खेल रही हैं|

Radha aur uski dost ki billiyan khel rahi hain.

Radha and her.GEN.SG friend GEN cat.PL play are.PL AUX.PL.PRS

Radha is playing with her friend's cats.

राधा और उसकी दोस्त की बिल्लियाँ खेल रही हैं|

Radha aur uski dost ki billiyan khel rahi hain.

Radha and her.GEN.SG friend GEN cat.PL play are.PL AUX.PL.PRS

Radha's cats are playing with her friend's cats.

राधा और उसकी दोस्त की बिल्लियाँ खेल रही हैं|

Radha aur uski dost ki billiyan khel rahi hain.

Radha and her.GEN.SG friend GEN cat.PL play are.PL AUX.PL.PRS

The cats belonging to Radha and her friend, are playing.

3. Semantically Ambiguous Sentences:

Sentence 5:

राधा ने अपनी दोस्त से कहा कि वह सुंदर है ।

Radha ne apnee dost se kaha ki wah sundar hai

Radha ERG GEN friend INS say.PST that DEM beautiful AUX.PRS.SG.

Radha told her friend that she (Radha) was beautiful.

राधा ने अपनी दोस्त से कहा कि वह सुंदर है ।

Radha ne apnee dost se kaha ki wah sundar hai

Radha ERG GEN friend INS say.PST that DEM beautiful AUX.PRS.SG.

Radha told her friend that she (her friend) was beautiful.

Sentence 6:

सब लोग कुछ खाओ!

Sab log kuch khao!

All people something eat.IMP!

All of you eat something (everyone eats the same thing)!

सब लोग कुछ खाओ!

Sab log kuch khao!

All people something eat.IMP!

All of you eat something (everyone eats different things)!

4. Pragmatically Ambiguous Sentences:

Sentence 7:

क्या तुम तैर सकते हो?

Kya tum taer sakte ho?

Q 2.SG swim can AUX.PRS.SG

Can you swim? (Question of ability)

क्या तुम तैर सकते हो?

Kya tum taer sakte ho?

Q 2.SG swim can AUX.PRS.SG

Can you swim? (Request)

Sentence 8:

राधा पुस्तक लेकर वहाँ चली |

Radha pustak lekar waha chali

Radha book take DEM go.PRS.F.SG

Radha went there with the book.

The place “There” is ambiguous in this sentence and depends on the context.

Part B:

1. हार की चमक मेरी आँखें में बर्करार थी।

GOOGLE TRANSLATE:

The glow of defeat remained in my eyes.

CHAT GPT:

The glow of defeat remained in my eyes.

The shine of the necklace was still reflected in my eyes.

2. ये आम हैं मगर फिर भी अच्छे हैं ।

GOOGLE TRANSLATE:

These are common but still good.

CHAT GPT:

These are ordinary, but still good.

These are mangoes, but still good.

3. वह साहसी कुत्ते का मालिक है।

GOOGLE TRANSLATE:

He is the owner of a courageous dog.

CHAT GPT:

He is the owner of a brave dog.

He is a brave owner of a dog.

4. राधा और उसकी दोस्त की बिल्लियाँ खेल रही हैं।

GOOGLE TRANSLATE:

Radha and her friend's cats are playing

CHAT GPT:

Radha's cats and her friend's cats are playing.

The cats belonging to Radha and her friend are playing.

Radha and (Radha's) friend's cats are playing.

5. राधा ने अपनी दोस्त से कहा कि वह सुंदर है ।

GOOGLE TRANSLATE:

Radha told her friend that she was beautiful.

CHAT GPT:

Radha told her friend that she (Radha) is beautiful.

Radha told her friend that she (the friend) is beautiful.

6. सब लोग कुछ खाओ!

GOOGLE TRANSLATE:

Everybody eat something!

CHAT GPT:

Everybody eat something!

Everyone, take some initiative / try something!

7. क्या तुम तैर सकते हो?

GOOGLE TRANSLATE:

Can you swim?

CHAT GPT:

Do you know how to swim?

Are you able to swim (right now)?

Do you think you can swim?
(implied challenge or suggestion)

8. राधा पुस्तक लेकर वहाँ चली।

GOOGLE TRANSLATE:

Radha went there with the book.

CHAT GPT:

Radha went there to take the book (or deliver the book)

Radha first took the book (from somewhere) and then went there.

Part C:

SENTENCE	TYPE OF AMBIGUITY	GOOGLE TRANSLATE	LLM EXPLANATION	TRANSLATION	OBSERVATIONS
हार की चमक मेरी आँखें मे बर्क़रार थी।	Lexical	The glow of defeat remained in my eyes.	The glow of defeat remained in my eyes. The shine of the necklace was still reflected in my eyes.	The sparkle of defeat remained in my eyes. The sparkle of the necklace remained in my eyes.	Lexical Ambiguity due to the two interpretations of हार : defeat and necklace
ये आम हैं मगर फिर भी अच्छे हैं।	Lexical	These are common but still good.	These are ordinary, but still good. These are mangoes, but still good.	These are mangoes yet they are still good. These are common yet they are still good.	Lexical Ambiguity due to the two interpretations of आम: Mangoes and Common/Ordinary
वह साहसी कुत्ते का मालिक है।	Syntactical	He is the owner of a courageous dog.	He is the owner of a brave dog. He is a brave owner of a dog.	He's the owner of the brave dog. He's the brave owner of the dog.	Syntactic Ambiguity in the attachment of साहसी: to the dog or to the owner.
राधा और उसकी दोस्त की बिल्लियाँ खेल रही हैं।	Syntactical	Radha and her friend's cats are playing	Radha's cats and her friend's cats are playing. The cats belonging to Radha and her friend are playing. Radha and (Radha's)	Radha is playing with her friend's cats. Radha's cats are playing with her friend's cats.	Syntactic Ambiguity in the attachment of Radha and her friend with respect to the cats and who owns or plays with them.

			friend's cats are playing.		
राधा ने अपनी दोस्त से कहा कि वह सुंदर है	Semantic	Radha told her friend that she was beautiful.	Radha told her friend that she (Radha) is beautiful. Radha told her friend that she (the friend) is beautiful.	Radha told her friend that she (Radha) was beautiful. Radha told her friend that she (her friend) was beautiful.	Semantic Ambiguity because of anaphoric ambiguity (वह can be traced back to Radha or her friend).
सब लोग कुछ खाओ!	Semantic	Everybody eat something!	Everybody eat something! Everyone, take some initiative / try something!	All of you eat something (everyone eats the same thing)! All of you eat something (everyone eats different things)!	Semantic Ambiguity because of quantifier scope ambiguity (everyone can eat the same thing or does everyone eat different things)
क्या तुम तैर सकते हो?	Pragmatic	Can you swim?	Do you know how to swim? Do you think you can swim? (implied challenge or suggestion)	Can you swim? (Question of ability) Can you swim? (Request)	Pragmatic Ambiguity as the sentence can be a request or a question of ability.
राधा पुस्तक लेकर वहाँ चली	Pragmatic	Radha went there with the book.	Radha went there to take the book (or deliver the book) Radha first took the book (from somewhere) and then went there.	Radha went there with the book. (there is ambiguous and depends in the context of the conversation)	Pragmatic Ambiguity as the demonstrative pronoun can have different interpretations based on the context present.

Part D:

Google Translate:

In all of the sentences, google translate **ignores the ambiguity** and **provides just one translation** for the Hindi phrases.

That's because Google Translate doesn't have the ability to analyze cultural context or access world knowledge.

Chat GPT:

GPT provides the different interpretations of the sentence in almost all of the cases. However, for lexical ambiguities, you have to specifically prompt it to look for **LEXICAL** ambiguities in the data. Without it, GPT starts talking of the emotional differences in sentences.

Eg. In हार की चमक मेरी आँखें मे बर्करार थी|,

GPT writes:

- Literal / Emotional Interpretation (most common)
“The sparkle of defeat lingered in my eyes.”
- Ironical / Paradoxical Interpretation
Even in defeat, there was a certain brilliance in my eyes
- In English, poetic versions might read:
“My eyes still reflected the glow of my loss.”
“Even in defeat, my eyes shone with a certain light.”

Instead of explaining the different meanings of हार.

However, in the sentence, राधा पुस्तक लेकर वहाँ चली| ambiguity resolution was **non-trivial**.

GPT says:

1. Radha went there to take the book (or deliver the book)

2. Radha first took the book (from somewhere) and then went there.

However, it doesn't shed light on the pragmatic ambiguity of 'there'.

Conculsion:

Google Translate shows a good understanding of the morphology of a language,

but **fails at understanding contextual cues and world knowledge.**

It provides only a single interpretation of the sentence, instead of explaining the different types of ambiguities.

Chat GPT on the other hand shows a clear understanding of:

1. **Context dependence:** Even without prior cultural and conversational context, GPT does a good job of listing all possible interpretations that might exist for a particular sentence.
2. **World knowledge:** GPT lists cases for क्या तुम तैर सकते हो?, understanding that the statement is often used as a question of ability and not always as a request.
3. **Lexical knowledge:** Though you have to prompt GPT to list the Lexical Ambiguities, it does a good job of understanding the different homonyms and polysemy present in the sentence.

All in all, **Chat GPT** succeeds in resolving cultural and pragmatic ambiguity to an appreciable degree.